MODEL	MD6310	MD6380
Hole Diameter	• 203-311 mm **(8-12.25 in)**	• 251 to 381 mm **(9.875-15 in)**
Hole Depth	**For short mast:** • Single-pass — 13.7 m **(44.9 ft)**, Multi-pass — Down to 74.6 m **(244.9 ft)** **For long mast:** • Single-pass — 17.5 m **(57.4 ft)**, Multi-pass — Down to 47.9 m **(157.4 ft)**	• Single-pass — 19.8 m **(65 ft)** • Multi-pass — Down to 39.6 m **(129 ft)**
Air compressor	• Rotary — 56.6 m^3/min **(2000 ft^3/min)** @ 8.6 bar **(125 psi)** • Rotary — 73.6 m^3/min **(2600 ft^3/min)** @ 7.6 bar **(110 psi)** • DTH — 42.2 m^3/min **(1500 ft^3/min)** @ 34.4 bar **(500 psi)**	• Rotary — 101.9 m^3/min **(3600 ft^3/min)** @ 6.9 bar **(100 psi)**
Engine	• C32 @ 1800 RPM • Emissions — U.S. EPA Tier 4 Final • Power Rating (ISO 14396) — 751 kW **(1007 hp)** • C32 @ 1800 RPM • Emissions — U.S. EPA Tier 2 equivalent • Power Rating (ISO 14396) — 769 kW **(1031 hp)**	• 3512C @ 1800 RPM • Emissions — U.S. EPA Tier 2 equivalent • Power Rating (ISO 14396) – 949 kW **(1273 hp)**
Cooler ambient rating	• Up to 52° C **(125° F)**	• Up to 52° C **(125° F)**
Cab	• FOPS cab has 3 m^2 **(32.4 ft^2)** floor space (standard cab) • Optional extended size cab for a total of 4.2 m^2 **(45.2 ft^2)** floor space	• FOPS cab has 4.2 m^2 **(45.2 ft2)** floor space
Fuel Tank	• Primary tank is 1609 L **(425 U.S. gal)** plus expansion of +1609 L **(+425 U.S. gal)** for total of 3218 L **(850 U.S. gal)** (option)	• 4164 L **(1100 U.S. gal)**
Pipe Rack	• 12.19 m **(40 ft)** pipes – for short mast • 7.62 m **(25 ft)** pipes – for long mast	• 9.9 m **(32.5 ft)** pipes
Drill Pipe Capacity	• Quantity — 4 or 5 pod	• Quantity — 2 pod
Tram Speed Maximum	• 2.45 km/h **(1.5 mph)**	• 1.93 km/h **(1.2 mph)**
Angle Hole Drilling	• Vertical to 30° (in 5° increments)	• Vertical to 30° (in 5° increments)

MODEL	MD6640
Hole Diameter	• 244-406 mm **(9.6-16 in)**
Hole Depth	• Single-pass — 21.3 m **(70 ft)** • Multi-pass — Down to 42.6 m **(140 ft)**
Air compressor	• 108 m^3/min **(3800 ft^3/min)** @ 4.5 bar **(65 psi)**
Engine	• Electric Motor • 597 kW **(800 hp)** • 4160V/7200V **(50 or 60 Hz)**
Cooler ambient rating	• Up to 52° C **(125° F)**
Cab	• FOPS cab has 4.2 m^2 **(45.2 ft^2)** floor space
Fuel Tank	• Not applicable
Pipe Rack	• 9.14 m **(30 ft)** or 12.19 m **(40 ft)** pipes
Drill Pipe Capacity	• Quantity — 2 pod
Tram Speed Maximum	• 1.77 km/h **(1.1 mph)**
Angle Hole Drilling	• Vertical to 25° in 5° increments

FOREST PRODUCTS

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Introduction

Cat forestry machines are specifically designed for tough forest work. Each model uses purpose built booms, sticks and grapples designed by Caterpillar for maximum performance and durability.

The following information provides features, specifications, dimensions, working ranges and major component weights for the forest machines.

Features

538/538 LL

- **Cat C7.1 ACERT engine** provides exceptional fuel efficiency while delivering maximum power for this machine size class.
- **Optimized hydraulics** minimize losses due to inefficiencies through component design, component layout and finely tuned machine and work tool parameters.
- **Performance** is enhanced with over 10% more swing torque, 12% more travel speed, and 10% more hydraulic flow compared to the previous model.
- **Rugged frame designs** are purpose built for forest applications; these include reinforcements to the upper frame, heavy duty doors and guarding, rugged swing bearing, reinforced carbody and roller frames that help protect your investment.
- **Undercarriage** is high-wide design for stability and ground clearance, and has 15% or 4" more ground clearance than the previous model. Undercarriage includes 203 mm (8 in) pitch 330 PPR3 track links; roller frames include eight bottom rollers total with full length track guards and high drawbar final drives.
- **Engineered reliability** is provided through the use of proven Cat components and increased cooling capacity with hydraulic reversing fan standard equipment and 2 A/C condensers for superior HVAC performance and cab cooling capability.
- **Diverse machine configurations** are available to meet the many applications of todays forest machines; these include front omission, road builder, rotate grapple, processor, under/under, over/under, powerclam, and material handler options.
- **Operator station** standard equipment includes heated and ventilated seat with lumbar support, 4 point seat belt, 1.25" or 32 mm polycarbonate front window, and high resolution 10" touchscreen display that integrates standard rear view camera and optional side view camera.
- **Diverse work tool options** include integrated heel racks and GLL grapples; both deliver high performance and reliability in severe logging applications.

558 LL

- **Cat C7.1 ACERT engine** provides exceptional fuel efficiency while delivering maximum power for this machine size class.
- **Optimized hydraulics** minimize losses due to inefficiencies through component design, component layout and finely tuned machine and work tool parameters.
- **Performance** is enhanced by 25% more swing torque than previous model that helps maximize work site efficiency.
- **Rugged frame designs** are purpose built for forest applications; these include reinforcements to the upper frame, heavy duty doors and guarding, rugged swing bearing, reinforced carbody and roller frames that help protect your investment.
- **Undercarriage** includes roller frames that deliver nine rollers for needed stability and to minimize point loading; 216 mm (8.5 in) pitch 349 HEX track are used for durability.
- **Engineered reliability** is provided through the use of proven Cat components and increased cooling capacity.
- **Diverse machine configurations** are available to meet the many applications of todays forest machines; these include under/under, over/under, processor, powerclam, and material handler options.
- **Operator station** consists of updated seat with heated and ventilated functionality along with a new improved LCD display that is 40% larger and includes four times the screen resolution; LED lights are standard on the machine. Cab options include side-entry cab and rear-entry cab on log loader configurations.
- **Diverse work tool options** include integrated heel racks and GLL grapples; both deliver high performance and reliability in severe logging applications.

568/568 LL

- **Cat C9.3 ACERT engine** provides exceptional power and fuel efficiency.
- **Attachments** — Factory installed log loader fronts with live heel and Cat grapples; harvester, road builder and butt-n-top fronts; help meet diverse forestry applications.
- **Cat GLL forestry grapples**, built with high-grade steel throughout the grapple, matched with Cat Forest Machines provide high performance and reliability in logging applications.
- **Reinforced carbody design** stands up to the most demanding forest applications, assuring outstanding durability and service life.
- **Undercarriage** — Heavy-duty link assemblies provide toughness and durability, maximizing undercarriage life and minimizing operating costs. Long 10 roller track frames provide excellent machine stability and maneuverability on steep slopes.
- **Guarding** — Purpose designed guarding helps extend service life, reduces downtime and helps protect your forestry machine investment.
- **Operator station** — Spacious, quiet, automatically climate controlled cab has excellent sightlines to the work area. Cat offers a premium rear entry cab for those regions or operations that require this functionality.
- **Serviceability** — Simplified service and maintenance, and electronic diagnostics help save time, money and increase productivity.

Wheel Loader Forestry Features:

Wheel loader Forestry-specific configurations equip the 990, 988, 980, 966, 962, 950, as well as XE models, with additional features to deliver the performance, strength and durability required to be productive in tough millyard applications including loading and unloading trucks, sorting, decking and feeding the mill.

Logging, millyard and wood pallet forks as well as woodchip and clean-up buckets can be added to equip the machines for forestry applications.

Ride control, axle oil coolers, heavy-duty tilt cylinders, factory 3V and 4V hydraulic options, lockup clutch, reversing fan, and additional counterweight are among some of the factory integrated solutions for use in forestry and logging applications.

Specially designed work tools for forestry applications are matched to the machines for optimum performance along with optional guarding packages for machine protection.

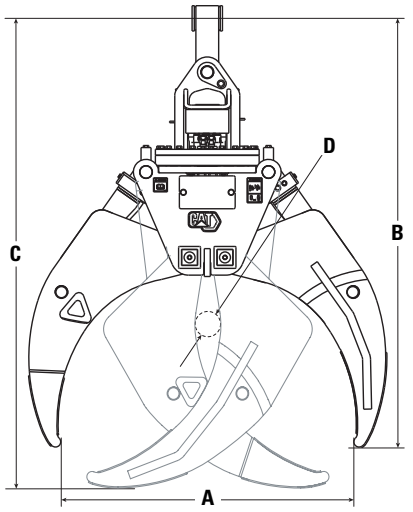
WORK TOOLS

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Features:

- **Full 360° continuous rotation**.
- **Paddle style tines** are made of high strength steel, and use abrasion resistant material on tips for longer life.
- **Induction hardened** pins and bushings.
- **Heavy duty hydraulic cylinders** with built in check valves, and protective hose guarding for more uptime.

Dimensions for Log Loading Grapples



	GLL52		GLL55		GLL60	
Part Number	**591-0042**		**591-0043**		**591-0044**	
Weight	1253 kg	**2,762.4 lb**	1297 kg	**2,860 lb**	1388 kg	**3,060 lb**
Width	561 mm	**22.09"**	561 mm	**22.09"**	561 mm	**22.09"**
A Maximum Opening	1321 mm	**52.01"**	1397 mm	**55"**	1524 mm	**60"**
B Height, Open	1839 mm	**72.4"**	1885 mm	**74.21"**	1975 mm	**77.76"**
C Height, Closed	1887 mm	**74.29"**	1958 mm	**77.09"**	2032 mm	**80"**
D Minimum Opening	127 mm	**5"**	127 mm	**5"**	127 mm	**5"**
Rotation, Continuous	**360°**		**360°**		**360°**	
Rotation, Torque	1153 N·m	**850.41 ft-lb**	1153 N·m	**850.41 ft-lb**	1153 N·m	**850.41 ft-lb**

Forestry Grapple Matching Guide

Optimal ● Acceptable ○ No match

Machine Model	GLL52	GLL55	GLL60
538 LL	●	○	
548 LL	●	●	
558 LL	○	●	●
568 LL	○	○	●
MH3040*		○	●
M325D L MH*		○	●
320 FM	●	○	
322 FM	●	●	
324 FM	●	●	
325 FM	○	●	●
330 FM	○	●	●

*When equipped with appropriate front linkage

Features:

Loader Fork

- Few work tool lines can match the range and utility of Cat Forks. Forks transform loaders into high performance material handling platforms capable of sorting, stacking and working wherever product, palletized material or lumber is at hand.

Millyard and Logging Forks

- Cat Millyard and double top clamp forks are performance-matched to Cat Wheel Loaders for unmatched on-the-job performance. The design features of fork and loader complement each other to make the ideal total system solution for log handling applications. Both forks are ideally suited for heavy-duty applications: loading and unloading trucks, sorting, decking, and feeding the mill.

Log and Lumber Forks

- Handle logs or finished lumber with equal ease. The top clamp holds loose loads securely, and the pallet-style forks make short palletized material. This versatility makes them suitable for a wide range of jobs including loading trucks, decking and sorting lumber or logs.

Features — Couplers

Fusion Couplers

- Fusion Quick Couplers provide unmatched versatility to any loader. Change tools in seconds without leaving the cab. Any work tool backed by coupler hooks can be picked up; allowing the loader to fit whatever application is at hand.

Center-Lock™ Pin Grabber Couplers

- The Cat Center-Lock Pin Grabber Coupler allows buckets and other standard work tools to be used without any modification. Exchanging work tools in seconds improves overall production and increases machine versatility. The Coupler is pinned on in place of the bucket with standard pins, and can be easily removed should the need arise to mount a tool directly to the stick.
- The Center-Lock Coupler offers new possibilities. Buckets can be turned around and used in front shovel mode for final trench clean up. An integral lift eye on the coupler body allows lifting without the weight of the bucket, increasing both machine capacity and visibility from the cab.

Features — Buckets and Thumbs

Cat Bucket Thumbs for Hydraulic Excavators

- Multiply the performance of a Cat Excavator by adding a Cat Bucket Thumb. This highly versatile work tool acts in conjunction with the bucket to allow the excavator to grab irregularly shaped items and load loose materials and debris.

Mini Bucket Thumbs

- Cat Bucket Thumbs are matched to Hydraulic Excavator Buckets for increased on-the-job performance. A thumb works with the bucket to grab, pick and sort debris, brush, trash and rock, opening up new production opportunities for your Cat Mini Excavator. Thumbs are an ideal complement to excavators working in demolition, land clearing, landscaping, material handling and construction jobs.

Clamshell Buckets

- Cat Clamshell Buckets are the premier tools for cleanup, demolition, ground clearing and forestry work. Built of heavy T1 steel, these buckets are tough and durable for long service life in the most difficult applications. Clamshell buckets feature continuous 360° rotation, powered by a high-torque hydraulic motor.

Features – Woodchip Dozers and Scoops

Buckets, U-blades, Bowldozers, Chip Scoops

- The high-capacity, high-efficiency design of these tools makes them high-production workhorses. They are matched to specific machines and material densities for optimum performance. The extra capacity and load retention capabilities ensure maximum usage and productivity.

Features — Rakes

Loader, Clearing and Clamp and Blade Rakes

- Rakes are durable, high-capacity tools that will increase production for land clearing, site cleanup and site preparation. Available in quick coupler and pin-on models, rakes pile brush, stack and carry debris and load trucks. Features include thick, fabricated teeth, a heavy-duty push bar and serrated tree pusher. A high brush rack retains the load, prevents back spillage and increases carrying capacity.

USE OF LOG VOLUME TABLES

The tabulated volumes on these pages were calculated with no taper in log diameter from base to top. Therefore each value listed in the table represents the volume of a true cylinder. In practice this may occur only in short sections of large diameter trees. To obtain the volume of solid wood logs, excluding bark:

1. Establish the base diameter of the log inside the bark and above the butt flare (extreme end taper).
2. Repeat the procedure for the top (small end) of log.
3. Enter log volume table at each of the two established diameters. Move horizontally to the vertical column closest to the length of the log being measured.
4. Establish the volume figures for each end of the log, add the two together and divide by two to obtain average log volume.

METRIC LOG VOLUMES (in Cubic Meters)

Log Diameter (cm)	LOG LENGTH (METERS)														
	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30
10	0.016	0.031	0.047	0.063	0.078	0.094	0.12	0.13	0.14	0.16	0.17	0.19	0.20	0.22	0.24
15	0.035	0.071	0.11	0.14	0.18	0.21	0.25	0.28	0.32	0.35	0.39	0.42	0.46	0.49	0.53
20	0.06	0.13	0.19	0.25	0.31	0.38	0.44	0.50	0.57	0.63	0.69	0.75	0.82	0.86	0.94
25	0.10	0.20	0.30	0.39	0.49	0.59	0.69	0.79	0.88	0.98	1.08	1.18	1.28	1.37	1.47
30	0.14	0.28	0.42	0.57	0.71	0.85	0.99	1.13	1.27	1.42	1.56	1.70	1.84	1.98	2.12
35	0.19	0.38	0.58	0.7	0.96	1.15	1.35	1.54	1.73	1.93	2.12	2.31	2.50	2.69	2.89
40	0.25	0.50	0.75	1.01	1.26	1.51	1.77	2.02	2.27	2.52	2.78	3.02	3.27	3.51	3.77
45	0.32	0.64	0.95	1.27	1.59	1.91	2.22	2.54	2.86	3.18	3.50	3.82	4.13	4.45	4.77
50	0.39	0.79	1.18	1.57	1.96	2.36	2.76	3.16	3.54	3.94	4.34	4.71	5.10	5.49	5.89
55	0.48	0.95	1.43	1.90	2.38	2.85	3.33	3.80	4.28	4.75	5.23	5.70	6.18	6.65	7.12
60	0.57	1.13	1.70	2.26	2.83	3.39	3.96	4.52	5.09	5.65	6.22	6.78	7.35	7.92	8.48
65	0.66	1.33	1.99	2.65	3.32	3.98	4.65	5.31	5.98	6.64	7.30	7.96	8.62	9.29	9.95
70	0.77	1.54	2.31	3.08	3.85	4.62	5.40	6.15	6.93	7.70	8.48	9.23	10.0	10.77	11.54
75	0.88	1.77	2.65	3.53	4.42	5.30	6.19	7.06	7.95	8.84	9.72	10.60	11.49	12.37	13.25
80	1.01	2.01	3.02	4.02	5.03	6.03	7.05	8.06	9.07	10.08	11.09	12.10	13.10	14.10	15.10
85	1.13	2.27	3.40	4.54	5.67	6.81	7.94	9.08	10.20	11.32	12.47	13.62	14.75	15.89	17.02
90	1.27	2.54	3.82	5.09	6.36	7.63	8.90	10.17	11.43	12.71	13.99	15.27	16.54	17.81	19.10
95	1.42	2.84	4.75	5.67	7.09	8.51	9.92	11.33	12.76	14.18	15.60	17.01	18.43	19.85	21.26
100	1.57	3.14	4.71	6.28	7.85	9.42	11.0	12.58	14.16	15.72	17.30	18.85	20.42	22.0	23.56
125	2.45	4.90	7.36	9.82	12.27	14.73	17.18	19.6	22.1	24.5	27.0	29.5	32.0	34.4	36.8
150	3.53	7.1	10.6	14.1	17.7	21.2	24.7	28.3	31.8	35.3	38.8	42.4	45.9	49.5	53.0
175	4.8	9.6	14.5	19.2	24.0	28.9	33.7	38.5	43.3	48.1	53.0	57.7	62.6	67.3	72.2
200	6.3	12.6	18.8	25.1	31.4	37.7	44.0	50.3	56.5	62.8	69.1	75.4	81.7	88.0	94.2

ENGLISH MEASURE LOG VOLUMES (in Cubic Feet)

Log Diameter (inches)	LOG LENGTH (FEET)																	
	8	**12**	**16**	**20**	**24**	**28**	**32**	**36**	**40**	**44**	**48**	**52**	**56**	**60**	**70**	**80**	**90**	**100**
4	0.7	1	1.4	1.7	2.1	2.4	2.8	3.1	3.5	3.8	4.2	4.5	4.9	5.2	6.1	7	7.8	8.7
6	1.6	2.4	3.1	3.9	4.7	5.5	6.3	7.1	7.8	8.6	9.4	10	11	12	13	16	18	20
8	2.8	4.2	5.6	7	8.4	9.8	11	13	14	15	17	18	19	21	24	28	31	35
10	4.4	6.5	8.7	11	13	15	17	20	22	24	26	28	31	33	38	44	49	55
12	6.3	9.4	13	16	19	22	25	28	31	35	38	41	44	47	55	63	71	79
14	8.5	13	17	21	26	30	34	39	43	47	51	56	60	64	74	86	96	101
16	11	17	22	28	34	39	45	50	56	61	67	73	78	84	98	112	126	140
18	14	21	28	35	42	49	57	64	71	78	85	92	99	106	124	141	159	177
20	17	26	35	44	52	61	70	79	87	96	105	113	122	131	153	175	196	218
22	21	32	42	53	63	74	85	95	106	116	127	137	148	158	185	211	238	264
24	25	38	50	63	75	88	101	113	126	138	151	163	176	189	220	251	283	314
26	29	44	59	74	89	103	118	113	147	162	177	192	207	221	258	295	332	369
28	34	51	68	86	103	120	137	154	171	188	205	222	240	256	299	342	385	428
30	39	59	79	98	118	137	157	177	196	216	236	255	275	295	344	393	442	491
32	45	67	89	118	134	156	179	201	223	246	268	290	313	335	391	447	503	559
34	50	76	101	126	151	177	202	227	252	277	303	328	353	378	441	504	567	631
36	57	85	113	141	170	198	226	255	282	311	339	368	396	424	495	566	637	707
38	63	95	126	158	189	220	252	284	315	347	378	410	441	473	551	630	709	788
40	70	105	140	175	210	244	279	314	349	384	419	454	489	524	611	698	785	873
50	109	164	218	273	327	382	436	491	545	600	645	709	764	818	955	1091	1227	1364
60	157	234	314	393	471	550	628	707	785	864	943	1021	1100	1178	1374	1571	1767	1964
70	214	321	428	535	642	748	855	962	1069	1176	1283	1389	1497	1604	1871	2138	2405	2673
80	279	420	559	698	838	977	1117	1257	1396	1536	1676	1815	1955	2095	2441	2293	3142	3491

WEIGHTS OF COMMERCIALLY IMPORTANT WOODS

Species	kg/m³ (Green)	lb/ft³ (Green)
A. Temperate Zone*		
Alder, Red	737	**46**
Ash, White	769	**48**
Aspen	689	**43**
Baldcypress	817	**51**
Basswood	673	**42**
Beech	865	**54**
Birch, Paper	801	**50**
Yellow	929	**58**
Cedar, Alaska	577	**36**
Incense	721	**45**
Northern, White	449	**28**
Port-Orford	897	**56**
Western Red	433	**27**
Cherry, Black	721	**45**
Cottonwood, Eastern	785	**49**
Douglas Fir, (Coast)	881	**55**
(Inland Empire)	577	**36**
Elm, American	865	**54**
Fir, Alpine	449	**28**
Balsam	721	**45**
Nobel	481	**30**
Red	769	**48**
Silver	577	**36**
White	753	**47**
Gum, Black	721	**45**
Blue	1121	**70**
Red	801	**50**
Tupelo	897	**56**
Hemlock, Eastern	801	**50**
Western	961	**60**
Hickory, Pecan	993	**62**
True	1009	**62**
Larch, Western	769	**48**
Locust, Black	929	**58**
Magnolia, Cucumber	785	**49**
Maple, Big Leaf	753	**47**
Black	865	**54**
Red	801	**50**
Silver	721	**45**
Sugar	897	**56**
Oak, Black	1009	**63**
Chestnut	977	**61**
Red	1009	**63**
Red, Swamp	1073	**67**
Swamp Chestnut	1041	**65**
White	993	**62**
White, Swamp	1105	**69**
Pine, Jack	801	**50**
Loblolly	993	**62**
Lodgepole	625	**39**
Long Leaf	993	**62**
Norway (Red)	673	**42**
Short Leaf	993	**62**
Slash	993	**62**
Sugar	817	**51**
Western Yellow, (Ponderosa)	721	**45**
White (Western)	561	**35**
White (Eastern)	577	**36**
Poplar, Yellow	609	**38**
Redwood	801	**50**
Spruce, Black	513	**32**
Engleman	625	**39**
Red	545	**34**
Sitka	529	**33**
White	545	**34**
Sweetgum	801	**50**
Sycamore	833	**52**
Tamarack	753	**47**
Walnut, Black	929	**58**
Willow, Black	801	**50**

***NOTE**: Weights taken from U.S. Dept. of Agriculture handbook No. 72, Wood Handbook.

Species	kg/m³ (Green)	lb/ft³ (Green)
B. Southeast Asia		
Apitong	961	**60**
Bintangor	865	**54**
Chumprak	929	**58**
Ebony	1746	**109**
Geronggang	721	**45**
Jelutong	641	**40**
Kapur (Borneo Camphorwood)	1073	**67**
Keruing	1121	**70**
Krabak	817	**51**
Kruen	1121	**70**
Lumbayau	929	**58**
Mahogany, Philippine		
(Red Luan)	753	**47**
(White Luan)	769	**48**
(Yellow Luan)	769	**48**
Mahoni	913	**57**
Alayan Kauri (Damar Minyak)	817	**51**
Melantai	705	**44**
Melapi	849	**53**
Mangkulang	929	**58**
Meranti Bakau	849	**53**
Meranti, Dark Red	753	**47**
White	769	**48**
Yellow	769	**48**
Mersawa	817	**51**
Nyatoh	897	**56**
Palosapis	817	**51**
Pulai	545	**34**
Ramin	1073	**67**
Rosewood (Sonokelina)	1314	**82**
Seraya, Dark Red	753	**47**
Yellow	769	**48**
White	769	**48**
Teak	1073	**67**

Species	kg/m³ (Green)	lb/ft³ (Green)
C. West Africa		
Abura	850	**53.06**
Ako	800	**49.94**
Azobe	1300	**81.16**
Aniegre (Mukali)	950	**59.31**
Bete	900	**56.19**
Bosse	900	**56.19**
Bubinga	1000	**62.43**
Dibetau	750	**46.82**
Douka (Makore)	950	**59.31**
Doussie	1200	**74.91**
Framire	850	**53.06**
Fromager	550	**34.34**
Ilomba	750	**46.82**
Iroko	1200	**74.91**
Kokrodua (Afrormosia)	1000	**62.43**
Kosipo	900	**56.19**
Limba	750	**46.82**
Mahogany	750	**46.82**
Moabi	1100	**68.67**
Niangon	900	**56.19**
Okoume	650	**40.57**
Ozigo	900	**56.19**
Padouk	1000	**62.43**
Samba (Obeche)	650	**40.58**
Sapelli	900	**56.19**
Sipo	800	**49.94**
Tchitola	850	**53.06**
Tiaba	900	**56.19**
Tola	850	**53.06**

Weights of Commercially Important Woods

- Australia
- New Zealand
- Papua New Guinea

Species	kg/m³ (Green)	lb/ft³ (Green)
D. Australia		
Ash Alpine	1041	**65**
Mountain	1009	**63**
Silvertop	1330	**83**
Black Butt	1121	**70**
Box Long Leaf	993	**62**
Yellow	1105	**69**
Black	1105	**69**
Brownbarrel	1073	**67**
Candle Bark	657	**41**
Gum Grey	1217	**76**
Manna	1121	**70**
Mountain	1169	**73**
Mountain Grey	1057	**66**
River Red	1137	**71**
Forest Red	1201	**75**
Southern Blue	1217	**76**
Spotted	1201	**75**
Sydney Blue	1153	**72**
Iron Bark Gray	1330	**83**
Narrowleaved	1330	**83**
Red	1330	**83**
Jarrah	1169	**73**
Karri	1169	**73**
Mahogany Red	1153	**72**
White	1282	**80**
Myrtle	1169	**73**
Peppermint	1120	**70**
Pine Radiata	865	**54**
Monerey	865	**54**
Celerytop	1057	**66**
Stringy Bark Brown	1233	**77**
Messmate	1169	**73**
Yellow	1217	**76**
White	1121	**70**
Tallowwood	1201	**75**
Wandoo	1282	**80**

Species	kg/m³ (Green)	lb/ft³ (Green)
E. New Zealand		
Exotic Softwoods		
Radiata Pine	1000	**62**
Douglas Fir	734	**45**
Corsican Pine	985	**61**
Redwood	1016	**63**
Larch	960	**60**
Indigenous Softwoods		
Mati	1120	**70**
Rimu	1130	**70**
Exotic Hardwoods		
Eucaliptus Botryoides	893	**56**
Eucaliptus Saligna	1200	**75**
Indigenous Hardwoods		
Beech — Silver	920	**57**
Beech — Red	1200	**75**
Tawa	1022	**64**

Species	kg/m³ (Green)	lb/ft³ (Green)
F. Papua New Guinea		
Pine, Hoop	520	**32**
Kauri	480	**30**
Klinki	510	**31**
Kwila	800	**50**
Erima	390	**24**
Taun	680	**42**
Walnut, PNG	560	**35**
Cedar, Pencil	720	**50**
Mersawa	650	**40**
Celtis, Hard	780	**48**
Rosewood, PNG	600	**37**
Beech, PNG	830	**51**
Oak, PNG	650	**40**
Ebony, PNG Black	1115	**69**
PNG White	720	**50**
Hardwood, Yellow	780	**48**
Hopea, Heavy	960	**60**
Light	710	**44**
Podocarp, Black	410	**25**
Terminalia, Brown	450	**28**

ESTIMATING NUMBER OF TREES PER HECTARE

Spacing (Meters)	Spacing (Meters)							
	1	2	3	4	5	6	7	8
1	10 000	5000	3333	2500	2000	1667	1428	1250
2	5000	2500	1667	1250	1000	834	714	625
3	3333	1667	1111	834	667	556	477	417
4	2500	1250	834	625	500	417	357	313
5	2000	1000	667	500	400	330	286	250
6	1667	834	556	417	333	278	238	208
7	1428	714	477	357	286	238	204	179
8	1250	625	417	313	250	208	179	156

ESTIMATING NUMBER OF TREES PER ACRE

Spacing (Feet)	Spacing (Feet)							
	5	6	7	8	9	10	11	12
5	1742	1452	1244	1089	968	871	792	726
6	1452	1210	1037	907	806	726	660	605
7	1244	1037	888	777	691	622	565	518
8	1089	907	777	680	605	544	495	453
9	968	806	691	605	537	484	440	403
10	871	726	622	544	484	435	396	363
11	792	660	565	495	440	396	360	330
12	726	605	518	453	403	363	330	302
13	671	558	478	418	372	335	304	279
14	622	518	444	390	346	311	283	259
15	580	484	415	363	323	290	264	242

COMPARISON OF LOG RULES ● Board Foot Values for 16-Foot Logs

Diameter at Small End, Inside Bark, Inches	International 1/4 Inch	Scribner	Scribner Decimal	Spaulding	Doyle
4	5	10	10	—	—
6	20	18	20	—	4
8	40	32	30	—	16
10	65	54	60	50	36
12	95	79	80	77	64
14	135	114	110	114	100
16	180	159	160	161	144
18	230	213	210	216	196
20	290	280	280	276	256
22	355	334	330	341	324
24	425	404	400	412	400
26	500	500	500	488	484
28	585	582	580	569	576
30	675	657	660	656	676
32	770	736	740	748	784
34	875	800	800	845	900
36	980	923	920	950	1024
38	1095	1068	1070	1064	1156
40	1220	1204	1200	1185	1296

UNIT OF MEASUREMENT DEFINITIONS

1 board foot	=	1/12 ft^3 of solid wood (1' × 1' × 1")
1000 board feet	=	83.33 ft^3 of solid wood
1 c. unit of wood	=	100 solid ft^3
	=	1200 board feet
	=	2.83^3
1 cord of wood	=	128 ft^3 of stacked logs
	=	3.62 m^3
1 unit of wood	=	200 ft^3 of loose chips
	=	5.66 m^3
1 cord of wood	=	0.85 units
1 Hoppus Ton	=	50 ft^3 (assumed)
	=	63.65 ft^3 (actual)
	=	600 board feet
	=	763.8 BF Brereton
	=	1.8 m^3 actual
	=	1.4 m^3 assumed
1 cubic meter	=	35.32 ft^3
	=	424 board feet
	=	333 board feet Hoppus tons
	=	0.555 Hoppus Tons
1 MBF Brereton	=	2.36 m^3
	=	785.4 board feet Hoppus
1 MBF Hoppus	=	1273 board feet-Brereton
MBF	=	Thousand board feet
1 Super Foot	=	1 board foot
100 Super Feet	=	1000 board foot
	=	0.236 m^3
600 Super Feet	=	50 ft^3
1 lb/ft^3	=	16.0185 kg/m^3

CUBIC FEET OF SOLID WOOD PER CORD

Length of Sticks-Ft.	Diameter at Small End 1"-2.5"	2.5"-5.5"	Over 5.5"
2	65	84	91
4	64	82	89
8	59	77	84
12	54	71	78

RULE OF THUMB CONVERSIONS

1 c. unit of wood = 1.117 cords = 1.25 units of chips = 250 ft^3 of chips = 7.08 m^3
1 cord of wood = 85 ft^3 of solid wood = 1.06 units of chips = 2.41 m^3
1 unit of chips = 80 ft^3 of solid wood = 2.27 m^3
1 cord of wood = 500 board feet = 1.18 m^3
2000 pounds of chips = 500 pounds of pulp
1 cord = 212 ft^3 of chips = 6 m^3

HYDRAULIC EXCAVATORS

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HYDRAULIC EXCAVATORS

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NEXT GEN MINI EXCAVATORS

The Next Generation Excavators have been imagined, designed, and built with an improved customer experience in mind. The work and development of this product has been done with a customer focused framework that was designed to exceed expectations when compared to the current Cat and competitive offerings. This product is truly differentiated from competitors and creates an overall experience in its size class.

To simplify the sales and marketing story for these products, an acronym has been developed for internal use to help make remembering areas of differentiation easy for dealer sales staff. The acronym, CAPES, highlights the five key areas of strength and differentiation of the products, and will be further detailed later in this document. See the overview below of the **CAPES** message:

C: Customer: These Next Gen Mini Excavators have been designed with the customer in mind and should exceed expectations with strong performance, differentiation, and the greatest operator experience. The safety of customers is at the forefront of the machine's design, seen in how the Next Gen Mini Excavators are equipped with a sealed and pressurized cab, and a three-inch hi-vis seatbelt, among other safety features.

A: Affordability: These Next Gen Mini Excavators, with their innovative, standard features, will provide a great price/value proposition for customers. It will also offer industry exclusive features like the canopy and tilt-up cab (some models), air conditioning (AC), the standard monitor with the option to upgrade to the touch-screen advanced monitor, and competitively priced basket of service parts.

P: Performance: These Next Gen Mini Excavators will excel versus competition and previous Cat models in productivity. The machines will also travel faster, dig to blade, and outperform any competitive model with regards to digging and trenching. In conjunction with Stick Steer, you can operate the machine just like a dozer with travel in the left joystick and the blade in the right joystick.

E: Experience: These Next Gen Mini Excavators will provide the most comfortable and unique customer experience of any excavators of its size. Some of the options include Stick Steer and Cruise Control, sealed and pressurized cab with AC, and a Next Gen LCD Monitor as standard with an option to upgrade to a Next Gen Advanced Monitor on cab units.

S: Serviceability: These Next Gen Mini Excavators boast an industry first and uniquely designed tilt-up cab that offers best in class access to all components and reduced service time. All common maintenance points are easily accessible through all-around service door access in addition to the tilt-up cab.

These Next Gen Mini Excavators will employ Cat Diesel engines that offer among the highest horsepower, displacement, and performance in the industry. The engine is Tier 4 Final Compliant and will offer the durability, performance, and serviceability that Cat engines are known for.

The Next Gen machines will feature nomenclature that corresponds to the weight of the machines in metric tons, just as the other machines in the Cat mini excavator lineup.

EXCAVATOR LIFTING CAPACITY

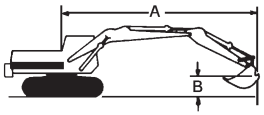
On many sewer jobs an excavator must lift and swing heavy pipe and manboxes in and out of the trench, place manholes and unload material from trucks. In some situations the excavator's lift requirements may be so critical that they determine the size excavator selected.

An excavator's lift capacity depends on its weight, center of gravity, the lift point position (see sketches) and its hydraulic capability. An excavator's lifting capability for any given lift position is limited by its tipping stability or hydraulic capacity.

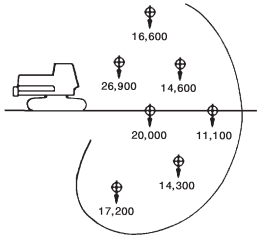
Changes in boom, stick and bucket position affect attachment geometry and can drastically change a machine's hydraulic lifting capacity. Caterpillar defines excavator lifting capabilities using the following SAE guidelines.

Tipping Conditions — An excavator is considered to be at the tipping point when the weight in the bucket acting at the center of gravity causes the rear rollers to lift clear of the track rails. Suspended loads are considered to be hung from the back of the excavator's bucket or bucket linkage by a sling or chain. Weights of attachments, slings or auxiliary lifting devices are considered part of the suspended load.

Thus, the tipping load is defined as the load producing a tipping condition at a specified radius. The load radius shall be measured as the horizontal distance from the axis of upper structure rotation (before loading) to the center of vertical load line with load applied (dimension A, below). The rating height is based on the vertical distance of the bucket lift point to the ground (dimension B).



A. Radius from swing centerline
B. Bucket lift point height



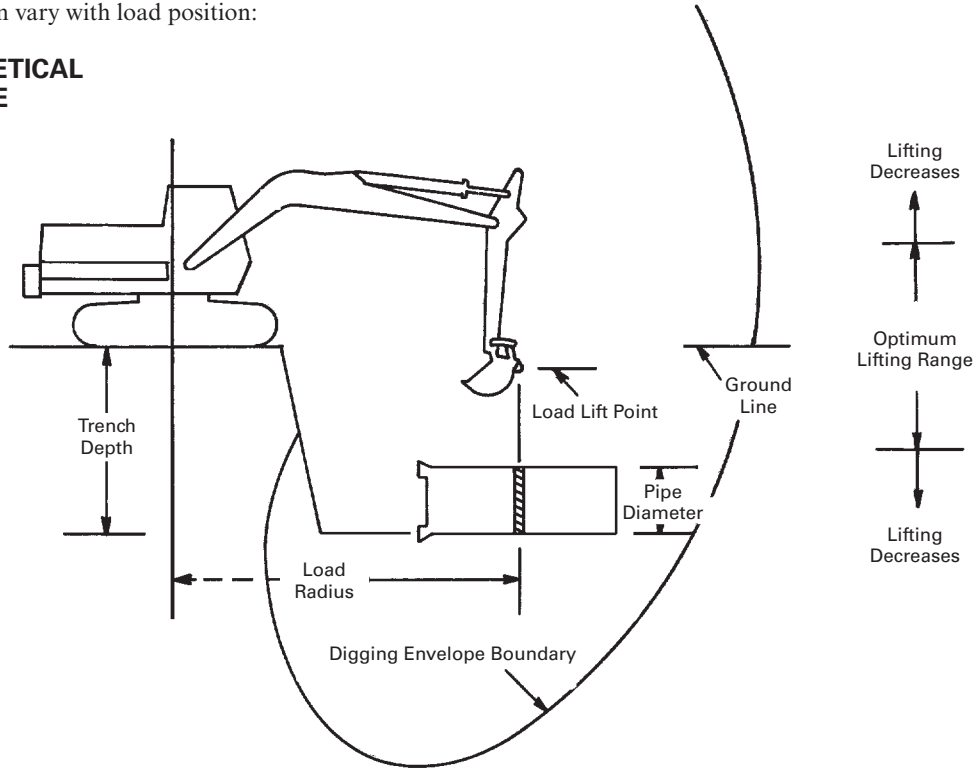
HYPOTHETICAL MACHINE

Rated Hoist Load — The rated load is established using the vertical distance of the lifting point to the ground and the radius of load. Ratings for the ability of a specific machine attachment to lift a load slung from the designated bucket are defined as follows:

a. The rated load will not exceed 75% of the tipping load.
b. The rated load will not exceed 87% of the excavator's hydraulic capacity. This means the machine should be able to lift 115% of the rated load.
c. The rated load will not exceed the machine's structural capability.

This drawing shows how an excavator's lifting capacity can vary with load position:

HYPOTHETICAL MACHINE



Tips for Lifting Above Ground:

Get the load as close to the excavator as possible.

Use a cable short enough and position the excavator so as to put the load lift point in the "optimum lifting range" (see sketch).

Problem: Long reach cable — Can't lift.

Solution: Shorten reach and cable — Can lift.

Tips for Lifting Below Grade:

Use a cable for sufficient length to position the load lift point in the "optimum lifting range".

Problem: Short cable, deep trench — Can't lift.

Solution: Lengthen cable to locate bucket hinge pin in optimum lifting area — Can lift.

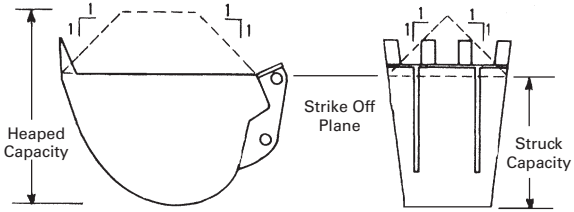
For lifting capacities at different heights please consult specific excavator model Technical Specifications.

EXCAVATOR BUCKET CAPACITIES

Caterpillar rates excavator buckets to conform with both PCSA standard No. 3 and SAE standard J-296. Buckets are rated on both their struck and heaped capacities as follows:

Struck Capacity

Volume actually enclosed inside the outline of the sideplates and rear and front bucket enclosures without any consideration for any material supported or carried by the spillplate or bucket teeth.



Excavator Bucket Rating

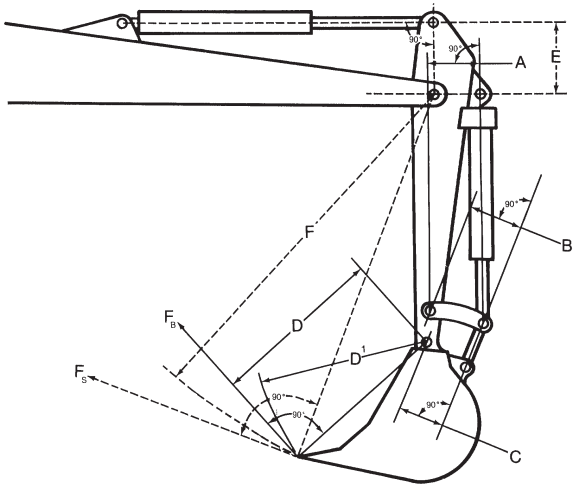
Heaped Capacity

Volume in the bucket under the strike off plane plus the volume of the heaped material above the strike off plane, having an angle of repose of 1:1 without any consideration for any material supported or carried by the spillplate or bucket teeth.

The Committee on European Construction Equipment (CECE) rates heaped bucket pay loads on a 2:1 angle of repose for material above the strike off plane.

CURL AND CROWD FORCES

Bucket penetration into a material is achieved by the bucket curling force (F_B) and stick crowd force (F_S). Rated digging forces are the digging forces that can be exerted at the outermost cutting point. These forces can be calculated by applying working relief hydraulic pressure to the cylinder(s) providing the digging force. The digging forces listed on next page conform with SAE Standard J1179 and PCSA Standard No. 3. The values may not be directly comparable to forces for machines rated by other methods than those described below.



F_B = Radial tooth force due to bucket cylinder

$$= \frac{\text{Bucket cylinder force}}{\text{Arm D length}} \left(\frac{\text{Arm A} \times \text{Arm C}}{\text{Arm B}} \right)$$

Cylinder force = (Pressure) × (End area of cylinder head)

Arm D = Bucket tip radius

Maximum radial tooth force due to bucket cylinder (bucket curling force) is the digging force generated by the bucket cylinder(s) and tangent to the arc of radius D^1. The bucket shall be positioned to obtain maximum output moment from the bucket cylinder(s) and connecting linkages. When calculating, maximum F_B occurs when the factor — Arm A times Arm C divided by Arm B — becomes the maximum.

F_S = Radial tooth force due to stick cylinder

$$= \frac{\text{(Stick cylinder force)} \times \text{(Arm E length)}}{\text{(Arm F length)}}$$

Arm F = Bucket tip radius + stick length

Maximum radial tooth force due to stick cylinder (stick crowd force) is the digging force generated by the stick cylinder(s) and tangent to the arc of radius F. The stick shall be positioned to obtain the maximum output moment from the arm cylinder and the bucket positioned as described in the bucket force rating. When calculating, maximum F_S occurs when the axis in the stick cylinder working direction is at a right angle to the line connecting the stick cylinder pin and the boom nose pin.

Bucket Selection Considering Bucket Curl and Stick Crowd Forces

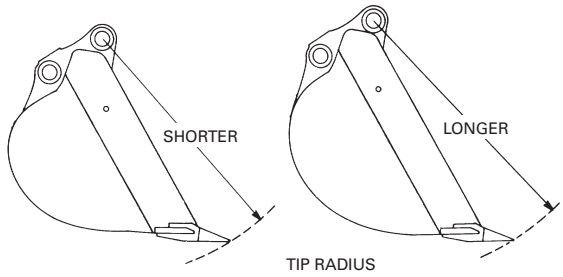
The combination of the excavator's stick crowd force and bucket curling force give this machine configuration more effective bucket penetration force per mm (inch) of bucket cutting edge than is available with other machine types such as wheel and track loaders.

As a result of high penetration force, an excavator bucket is comparatively easy to load. Also, the higher unit breakout forces allow the excavator's economic application range to be extended farther into the tougher soils (coral, caliche, shale, limestone) before blasting or ripping is required.

To take full advantage of an excavator's high penetration forces, buckets should be selected so they are well matched to soil conditions that are encountered. The two important things to consider are bucket width and bucket tip radius.

As a general rule, wide buckets are used in easily dug soil and narrow buckets in harder material. In hard rocky soils, tip radius also has to be considered in bucket selection. Because the shorter tip radius buckets provide more total bucket curling force than the long tip radius buckets, they are generally the easiest to load. A good rule of thumb when selecting a Cat bucket for hard material is to choose the narrowest bucket that has a short tip radius.

Other factors such as trench bottom width specifications, manbox size, or the desire to conserve bedding material may also influence excavator bucket selection.



Caterpillar offers a very comprehensive list of high strength steel buckets. High strength steel allows thinner components which helps keep the weight down, maintains durability and improves loadability. The wrong bucket can reduce production 30-40% or more. Caterpillar's in-depth knowledge of machine design, bucket design and application experience allows offering **machine matched** packages that optimize performance.

Additional buckets may be available and the listed buckets may not be available in all sales areas. Contact your Cat dealer for your specific bucket needs.

NOMENCLATURE FOR HYDRAULIC EXCAVATOR BUCKETS

Caterpillar has moved to a global nomenclature for buckets on small, medium and large excavator (HEX) machines and associated linkages. The end result has been a manageable, consistent bucket offering in all regions regardless of where the machine or bucket may be built and shipped.

Why the change?

Consistent product line

Buckets sold into different regions of the world have been designed and manufactured independent of each other resulting in different bucket styles and nomenclature. As Caterpillar became more global, we recognized this lead to confusion in the marketplace as machines and buckets are sourced from different areas.

Improved bucket selection

Caterpillar continues to recognize the need for a range of buckets for various applications and with varying durability requirements — from site development to granite quarries. The new, global nomenclature is simplified, consistent, and based on the durability of the bucket. These considerations allow a clear and consistent bucket positioning to help facilitate appropriate bucket selection and recommendation — and put Caterpillar in a better position to support machines on a global scale.

In the past, HEX buckets were grouped into three main categories: General Purpose (GP), Heavy Duty (HD), and Heavy Duty Rock (HDR). New nomenclature brings in four primary categories, each represent durability. They are: General Duty (GD), Heavy Duty (HD), Severe Duty (SD), and Extreme Duty (XD). Within the Extreme Duty class a new bucket specific for use in granite will be available for the large excavators.